Technical Details of the Annotation Extender

1 Pre-defined Tagset Mapping Schema

The annotation extension process is based on a **Pre-defined Tagset Mapping Schema**, which functions as a linguist-designed, language-specific reference guiding annotation decisions. This schema is prepared prior to the implementation of the extension procedure and is not directly used as a system input. Instead, it specifies, for each error type, (i) whether meaningful facet-level information can be inferred and (ii) if so, which inference strategy (direct or context-dependent) should be applied. In this way, the framework is adapted to the linguistic characteristics of different error types and allows for language-specific customization, since the schema can be modified for other languages without requiring changes to the core system architecture.

Table 1 presents the Pre-defined Tagset Mapping Schema for Turkish, which consists of 37 error types and systematically links the error categories defined in the target learner corpus to the faceted taxonomy. The first two columns list the error types and their IDs as they are represented in the corpus-specific tagset, while the last six columns correspond to the linguistic facets/subfacets defined in the faceted taxonomy. The Error Rate column quantifies the prevalence of the respective error type within the entire corpus. It should be noted that this column is provided for informational purposes; it does not formally belong to the tagset mapping schema itself.

An examination of the table reveals four distinct types of value representations: (i) placeholders (e.g., `<upos>`, `<uif>`, `<ulf>`), (ii) single-valued entries (e.g., *Punct* in UPOS; *Typo* in ULF), (iii) multi-valued entries (e.g., *Lemma, Affix* in UNIT; *A, O, M* in the PHENOMENON), and (iv) dash symbols (–). In practical terms, their interpretation is as follows:

(i) **Placeholders** indicate that the facet value must be selected from the relevant UD value inventory. These values are inferred automatically through morphological analysis using UDPipe. For example, `<upos>` corresponds to obtaining a value from the POS tagset defined in UD.

(ii) **Single-valued entries** indicate that the facet consistently receives the same predefined value for the corresponding error type. In such cases, inference does not involve selection among alternatives; instead, the extender assigns this value deterministically. For example, in the case of the *Vowel Harmony* error type, the UNIT facet is always realized as *Affix*.

(iii) **Multi-valued entries** indicate that the facet takes a single value from a fixed set of allowed options, requiring context-aware value selection during inference.

(iv) **Dash symbols** (-) indicate that the facet is not linguistically meaningful for the corresponding error type and therefore need not to be inferred.

These four value types, together with the specific error type indicated by the column ERROR ID, encode how the corresponding facet should be handled during the annotation extension process. In other words, this tabular structure directly informs the inference strategy in the extender implementation, indicating when a facet value must be inferred, when it should be assigned deterministically, when value selection is required, and when the facet is not meaningful for a given error type.

Table 1: Pre-defined Tagset Mapping Schema for Turkish. UPOS: Universal Parts-of-Speech, UIF: Universal Inflectional Features, ULF: Universal Lexical Features, A: Addition, O: Omission, M: Misuse, MO: Misordering, Amb: Ambiguity. DIG error type is out of scope for the extension process because these errors are not generated by the learners themselves.

ERROR	ERROR ID	ERROR RATE(%)	UPOS	UIF	ULF	UNIT	PHENOMENON	LEVEL
Punctuation	PUN	**13.27**	Punct	–	–	Word, Sentence	A, O, M, MO	Orthography
Spelling	SPE	9.47	<upos>	<uif>	Typo	Lemma, Affix	A, O, M, MO	Orthography
Spacing	SPA	4.57	<upos>	<uif>	<ulf>	Affix, Word	A, O	Orthography
Capitalization	CAP	6.39	<upos>	–	<ulf>	Word, Phrase, Sentence	M	Orthography
Diacritics	DIA	7.95	<upos>	–	<ulf>	Grapheme	A, O, M	Orthography
Abbreviation	ABB	0.02	<upos>	–	Abbr	Word	M	Orthography
Consonant Voicing	CV	0.16	<upos>	<uif>	<ulf>	Lemma, Affix	M	Morphophonology
Vowel Dropping	VD	0.15	<upos>	–	<ulf>	Lemma	M	Morphophonology
Vowel Harmony	VH	3.22	<upos>	<uif>	<ulf>	Affix	M	Morphophonology
Buffer Letters	BL	0.25	<upos>	<uif>	<ulf>	Affix	A, O, M	Morphophonology
Consonant Assimilation	CA	0.43	<upos>	<uif>	<ulf>	Affix	M	Morphophonology
Vowel Narrowing	VN	0.06	<upos>	–	<ulf>	Lemma	M	Morphophonology
Consonant Doubling	CD	0.06	<upos>	–	<ulf>	Lemma	M	Morphophonology
Sequencing	SEQ	2.23	–	–	–	Affix, Phrase, Sentence	MO	Grammar
Case	CASE	**12.62**	<upos>	<uif>	<ulf>	Affix	A, O, M	Grammar
Number	NUM	2.06	<upos>	<uif>	<ulf>	Affix	A, O	Grammar
Possession	POSS	4.57	<upos>	<uif>	<ulf>	Affix	A, O, M	Grammar
Voice	VOICE	1.13	<upos>	<uif>	–	Affix	A, O, M	Grammar
Tense	TENSE	1.70	<upos>	<uif>	–	Affix	M	Grammar
Mood	MOOD	1.65	<upos>	<uif>	–	Affix	A, O, M	Grammar
Aspect	ASP	0.34	<upos>	<uif>	–	Affix	M	Grammar
Negation	NEG	0.19	<upos>	<uif>	–	Affix, Word	A, O, M	Grammar
Person	PER	1.41	<upos>	<uif>	–	Affix	M	Grammar
Non-Finite Verb	NFV	2.48	<upos>	<uif>	–	Affix	M	Grammar
Interrogative Particle	IP	0.07	–	–	–	Affix	A, O	Grammar
Ki Usage	KU	0.44	<upos>	–	–	Affix	A, O	Grammar
Unnecessary Affix	UA	0.42	<upos>	–	–	Affix	A	Grammar
Lexical Category	LC	1.20	<upos>	–	–	Word	M	Grammar

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Table 1 – continued from previous page

ERROR	ERROR ID	ERROR RATE (%)	UPOS	UIF	ULF	UNIT	PHENOMENON	LEVEL
Derivation	DE	0.18	<upos>	–	–	Affix	A, O, M	Grammar
Allomorphy	ALL	0.30	<upos>	<uif>	–	Affix	M	Grammar
Final-Initial Merge	FIM	0.29	Verb	<uif>	–	Word	O	Grammar
Descriptive Compound Verb	DCV	0.07	<upos>	<uif>	–	Affix	O	Grammar
Word Error	WE	**17.12**	–	<uif>	<ulf>	Word, Phrase	A, O, M	–
Awkward Phrasing	AP	1.54	–	–	–	Phrase, Sentence	M	–
Unclear Meaning	UM	0.06	–	–	–	Word, Phrase, Sentence	Amb.	Semantics
Style	STYLE	0.97	–	–	–	Affix, Word	M	Pragmatics
Digitization	DIG	0.96	–	–	–	–	–	–

2 System Architecture

Figure 1 presents the complete process for the annotation extender. In this study, the JSON export option of Label Studio was selected, as it preserves the full annotation structure (including task and annotation identifiers, span boundaries, error types, corrected forms, etc.) in a machine-readable hierarchical format.

The process begins with a Data Validation stage, which verifies compliance with the annotation principles established prior to the annotation process, such as requiring annotators to provide an explicit corrected form for each annotated error, permitting nested span annotations, disallowing partially overlapping spans.

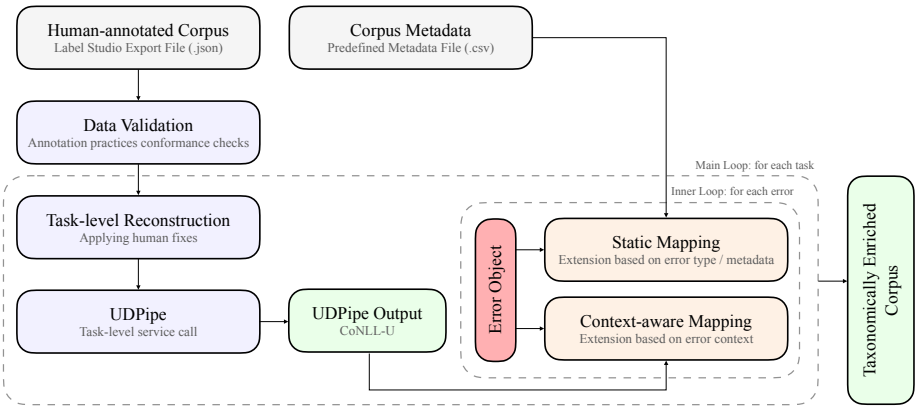


Figure 1 System Architecture of the Annotation Extender

Following validation, the system enters a *task-level main loop* that processes each annotated unit independently. The term "task" is coming from Label Studio terminology and for our annotation setup in Label Studio it corresponds to a single learner's essay. Within this loop, the Task-level Reconstruction component applies the human-provided corrected forms to the learner text in order to produce an error-free version of the essay. This reconstructed text constitutes a task-level target hypothesis, automatically composed from the span-based corrections supplied during human annotation; reconstruction is necessary because the morphological analysis used to infer the facet values is context-dependent and therefore requires each corrected token to appear within a well-formed sentence.

The reconstructed essay is then processed through the UDPipe to get the morphological analysis results. In this study, UDPipe v2 is employed with the `turkish-imst-ud-2.15-241121` (Sulubacak et al., 2016) pretrained model. The CoNLL-U formatted result is stored for use in the extension process.

Next, the system enters an *error-level inner loop* that iterates over each error instance. For every instance, an `Error` object is created, and its core attributes (such

as error type, span indices, corrected form, task identifier) are extracted directly from parsed Label Studio export file. This object is then passed to the extension module, where facet/subfacet values are assigned based on the Pre-defined Tagset Mapping Schema. The extension process integrates multiple inference strategies: (i) *static mapping* used for corpus metadata and single-valued schema entries; (ii) *context-aware mapping* used for selection of multi-valued schema entries, and for inferring *POS*, *InflectionalFeature*, and *LexicalFeature* subfacet values, which have placeholder entries in the schema, using CoNLL-U output parsing. **Static Mapping** is performed for pre-determined facet values that do not require any form of value selection or context-sensitive inference. That is, if a facet or subfacet value is fixed and remains the same across all instances of a particular error type, the extender directly assigns this value without consulting additional linguistic evidence. **Context-aware Mapping** is used when error-type information alone is insufficient to determine the correct facet or subfacet values. While some facets/subfacets can be inferred through static mappings (e.g., single-valued schema entries, subfacets for *Metadata*), others require linguistic evidence drawn from the local context of the error instance.

The example in Figure 2 shows how the three enrichment strategies work together to populate the facets of a single error. Although the figure presents only the final annotation, the facet values are obtained through different procedures. First, UDPipe provides the morphological information. For the corrected form *başarılı* ('successful'), UDPipe identifies the part of speech as `ADJ`, which is assigned to the POS subfacet. The InflectionalFeature subfacet remains empty because the adjective does not contain any overt inflectional marking. Second, *static mapping* assigns facet values that are fixed for all spelling errors. Regardless of the specific token, every error labelled *Spelling* receives the values `LexicalFeature=Typo` and `LinguisticLevel=Orthography`. These values therefore do not need to be inferred from the individual error instance. Third, *context-aware mapping* determines the facet values that depend on the specific error. In the example *başarlı* → *başarılı*, the missing vowel *ı* occurs within the word stem rather than in a suffix. The error is therefore assigned *Unit* = *Lemma*. The system then compares the erroneous and corrected forms at the character level and detects that a character is missing. Based on this comparison, the error is assigned *Phenomenon* = *Omission*. Taken together, this example shows how a single error is enriched through three complementary processes: morphological information supplied by UDPipe, fixed assignments based on the error type, and context-sensitive decisions derived from the relationship between the erroneous and corrected forms.

To illustrate the role of UDPipe, consider the corrected word *başarılı* ('successful') in the reconstructed phrase "... ve başarılı olmak için ...", derived from the spelling error shown in Figure 2. UDPipe analyses the corrected sentence and produces a linguistic description for each word. Among other information, it identifies the word's lemma (dictionary form), part of speech, and morphological features. For *başarılı*, UDPipe returns the information shown below.

FORM	LEMMA	UPOS	XPOS	FEATS
başarılı	başarı	ADJ	Adj	_

Manual Annotation

Label Studio


```
{
  "value": {
    "start": 643,
    "end": 650,
    "text": "başarlı",
    "labels": ["SPE"]
  },
  "id": "45ID60k62-",
  "type": "labels",
},
{
  "value": {
    "text": ["başarılı"]
  },
  "id": "45ID60k62-",
  "type": "textarea"
}
```


Taxonomically Enriched Annotation

Semi-automatic Annotation Extender


```
{
    "id": "45ID60k62-",
    "idLabelStudio": 13846,
    "idData": 658,
    "rawText": "Başarı olmak …",
    "idxStartErr": 643,
    "idxEndErr": 650,
    "incorrText": "başarlı",
    "errType": "SPE",
    "corrText": "başarılı",
    "errTax": {
      "id": "45ID60k62-",
      "pos": [{
            "form": "başarılı",
            "pos": "ADJ",
            "xpos": "Adj"
          }],
      "infFeat": [{
            "form": "başarılı",
            "feats": ""
          }],
      "lexFeat": [{
            "form": "başarılı",
            "feats": "Typo=Yes"
          }],
      "unit": "Lemma",
      "phenomenon": "Omission",
      "level": "Orthography"
    },
    "metadata": {
      "id": 658,
      "nationality": "palestine",
      "gender": "male",
      "topic": "success and pride"
    }
}
```


Figure 2 Output comparison of the original learner corpus annotation using Label Studio (left-side) and the taxonomically enriched annotation produced by the annotation extender. Attributes that were automatically added by Label Studio as part of its internal operating mechanism were excluded from the JSON representation in the Manual Annotation part. The example error is **başarlı** → **başarılı** ('successful'), a spelling error in which the vowel 'ı' is omitted, and is annotated as a Spelling (SPE) error. In the Label Studio export, each error is represented by two linked entries that share the same id: a **labels** entry, which records the error span (`"text": "başarlı"`) and its type (`SPE`), and a **textarea** entry, which records the annotator's corrected form. The corrected form thus appears in the textarea entry as `"text": ["başarılı"]`.

From this output, the extender automatically fills two subfacets. The part-of-speech tag (`ADJ`) is assigned to the POS subfacet. The morphological features field (`FEATS`) is used to populate the InflectionalFeature subfacet.

In this example, the field is empty (`_`) because the adjective *başarılı* contains no overt inflectional marking. In other words, UDPipe automatically provides the

grammatical information represented by the placeholders in the Pre-defined Tagset Mapping Schema (Table 1), removing the need for annotators to enter these values manually.

3 Evaluation

To assess the performance of the annotation extender, a gold-standard reference was first established through a controlled manual extension process. The original manual annotation serves as the input to the extender, not as a reference for its output: each annotation consists only of an error span, an error type, and a correction, and does not contain the facet values (POS, inflectional and lexical features, Unit, and Phenomenon) that the extender infers. Evaluating the extender therefore requires a reference for precisely these facet values, which the original annotation does not provide.

Stratified sampling strategy based on error type frequencies was used to create a balanced evaluation subset. The natural distribution of the corpus was preserved, but sampling quotas were changed to maximize the representation of rare errors and decrease the dominance of relatively frequent ones. Formally, given an error type et with frequency $\text{count}(et)$, the sampling size $s(et)$ was determined as:

$$s(et) = \begin{cases} \text{count}(et), & \text{if } \text{count}(et) \leq 30 \\ \max(30, \lceil 0.05 \times \text{count}(et) \rceil), & \text{otherwise} \end{cases}$$

All instances of low-frequency error types (≤ 30 occurrences) were fully retained, while for high-frequency types, approximately 5% of instances were sampled with a minimum threshold of 30 to ensure analytical reliability. Random sampling without replacement was then applied within each error-type stratum. This design increases the coverage of low-frequency errors and limits the dominance of highly frequent categories; however, it also deviates from the corpus' natural error distribution, and therefore primarily supports balanced diagnostic analysis rather than a fully distribution-representative performance estimate.

Following the computation and aggregation of error frequencies, 2,024 error instances were selected as an evaluation subset for human annotation. This selection was conducted via stratified sampling across 36 distinct error types, with the "DIG" category excluded from the analysis.

Two annotators independently enriched all error instances in the evaluation subset for all linguistic facets/subfacets, except for the *LinguisticLevel*, which was deterministically assigned via the predefined tagset mapping schema. Cohen's Kappa was used to evaluate inter-annotator agreement at the error-instance level, considering any difference between facets to be a complete disagreement. Significant agreement is indicated by the ensuing κ score of 0.80. Given differences in annotator experience, the senior expert's annotations were adopted as the ground truth for evaluating the extender's output.

As illustrated in Table 2, the annotation extender achieved high facet-level accuracy across various linguistic dimensions, reaching a **macro-average accuracy of 95.86%**. Performance in the *POS*, *InflectionalFeature*, and *LexicalFeature* subfacets

Table 2 Facet-based accuracy results for the annotation extender.

Facets/Subfacets	**Facet-level Accuracy (%)**
Part-of-Speech	94.46
Inflectional Feature	98.91
Lexical Feature	99.85
Unit	96.29
Phenomenon	89.77
Macro-average	**95.86**

is primarily a reflection of UDPipe's accuracy. Conversely, the *Unit* and *Phenomenon* facets—which rely on context-based heuristic inference—exhibit slightly lower performance. These results suggest that the extender generates linguistically consistent annotations that align closely with expert judgments across the evaluated error categories.

Additionally, the system's performance was measured by **annotation-level accuracy**, defined as the frequency of exact matches across all facet values relative to the gold standard. A total of 1,643 instances (81.18% of the 2,024 evaluated) were identified as exact matches. These findings indicate that the extender can reproduce comprehensive facet configurations for many error instances.

References

Sulubacak, U., Gokirmak, M., Tyers, F., Çöltekin, Ç., Nivre, J., Eryiğit, G.: Universal Dependencies for Turkish. In: Matsumoto, Y., Prasad, R. (eds.) Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, pp. 3444–3454. The COLING 2016 Organizing Committee, Osaka, Japan (2016). https://aclanthology.org/C16-1325/